

Queue ADT, Operations & Array Implementation

Q. Define Queue as an Abstract Data Type (ADT). Explain its standard operations and demonstrate its implementation using an Array with a relevant code snippet.

> **Definition to Remember**

> A **Queue** is a linear data structure that follows the First In First Out (FIFO) principle. Elements are added at the rear and removed from the front. It is a type of ADT that restricts the operations to two specific ends: enqueue (add) and dequeue (remove).

1. Queue as an ADT

As an Abstract Data Type, a Queue manages a collection of elements and restricts access to two specific ends:

- **Enqueue:** Adding an element to the rear of the queue.

2. Standard Operations

Operation	Description	Condition
Enqueue	Add an element to the rear of the queue.	Queue is not full.
Dequeue	Remove an element from the front of the queue.	Queue is not empty.

3. Array Implementation in C

Two pointers (`front` and `rear`) are initialized to `-1`.

```
```c
#include <stdio.h>
#define SIZE 100

int queue[SIZE];
int front = -1;
int rear = -1;

void enqueue(int value) {
 if (rear == SIZE - 1) {
 printf("Queue is full\n");
 return;
 }
 rear++;
 queue[rear] = value;
 printf("Element %d is enqueued\n", value);
}

int dequeue() {
 if (front == -1) {
 printf("Queue is empty\n");
 return -1;
 }
 int value = queue[front];
 front++;
 printf("Element %d is dequeued\n", value);
 return value;
}

int main() {
 enqueue(10);
 enqueue(20);
 enqueue(30);
 dequeue();
 dequeue();
 return 0;
}
```
```

4. Advantages and Limitations

- **Advantage:** Array implementation is simple and fast ($O(1)$ time complexity for enqueue/dequeue).

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

Complete Executable C Code: Linear Queue Array Implementation

```
```c
```

```
#define MAX 5
```

```
struct Queue {
```

```
void initQueue(struct Queue *q) {
```

```
int isFull(struct Queue *q) {
```

```
int isEmpty(struct Queue *q) {
```

```
return
```

```
void enqueue(struct Queue *q, int val) {
```

```
if (isFu
```

```
int dequeue(struct Queue *q) {
```

```
if (isEr
```

```
void display(struct Queue *q) {
```

```
if (isEr
```

```
int main() {
```

```
struct
```

```
enqueue(&q, 10);
```

```
enque
```

```
printf("Dequeued: %d\n", dequeue(&q));
```

```
display
```

```
return 0;
```

```
}
```